TAG NUMBER		39-HA-901 A/B
SERVICE		CARGO RE-HEATER (2 X 100%)
DUTY CASE DUTY	kW	3000
SHELL ID X T/T LENGTH	mm	610 X 2650
SHELL DESIGN / OPER.PRESS.	barg	FV & 16.5 / 3.0
SHELL DESIGN / OPER.TEMP.	degC	-10 & 280 / 30 & 40
TUBE DESIGN / OPER.PRESS.	barg	FV & 16.5 / 4.5
TUBE DESIGN / OPER.TEMP.	degC	-10 & 280 / 250
MAXIMUM FLOW HOT SIDE	kg/h	
MAXIMUM FLOW COLD SIDE	kg/h	
MATERIAL		DSS
VENDOR		STF LOTERIOS
EQUIPMENT PACKAGE		ER307

NOTES

1. TEMPERATURE TRANSMITTERS TO BE MOUNTED AS CLOSE AS POSSIBLE TO HEATER OUTLET FLANGE.
2. FOR PORTABLE PUMP AND HOSE CONNECTION.
3. DELETED.
4. DELETED.
5. THE VALVE (41-XSV-8501) SHALL BE CLOSED WHEN HIGH-HIGH OF 39-TST-9515 OR 39-TST-9535 TO BE LOCATED AT HIGH POINT LOCAL TO CARGO RE-HEATER.
6. 39-TV-9514 IS NOT ALLOWED TO OPEN BEFORE 39-HV-9511 IS CONFIRMED OPEN AND CONFIRMED MINIMUM FLOW ON 39-FIT-9503A.
   39-TV-9534 IS NOT ALLOWED TO OPEN BEFORE 39-HV-9531 IS CONFIRMED OPEN AND CONFIRMED MINIMUM FLOW ON 39-FIT-9503A.
7. A START UP PERMISSIVE IS REQUIRED TO RESET AND OPEN 41-XSV-8501.
   39-TV-9514 AND 39-TV-9534 BOTH MUST BE CLOSED TO ALLOW OPENING OF 41-XSV-8501.
8. NUMBER AND LOCATION OF LOW POINT DRAINS BASED ON PIPING LAYOUT.
9. PERSONNEL PROTECTION INCLUDED FOR SECTIONS OF THE PIPE WHERE PERSONNEL MIGHT BE EXPOSED, AS PER THE INSTALATION PHILOSOPHY.
10. VALVE (41-XSV-8501) SHALL BE CLOSE AT HIGH-HIGH POINT AND LOW-LOW POINT OF 39-FIT-9503B.
11. SAFETY CRITICAL HEAT TRACING - WAX MITIGATION (65°C).
12. WASH WATER LINE FOR TANK CLEANING.
13. FLOW METER STRAIGHT LENGTH REQUIREMENT - 5D UPSTREAM AND 3D DOWNSTREAM.
14. 39-BU-9501 TO BE INTERLOCKED WITH 39-BU-9888 SUCH THAT ONLY ONE VALVE IS OPEN AT A TIME.
15. SELECTOR SWITCH PROVIDED TO SELECT MODE OF OPERATION FOR CONTROL VALVE (39-FV-9503A). FLOW CONTROLLER (39-FIC-9503A) SHALL BE SELECTED DURING CRUDE OIL TANK TO TANK TRANSFER OPERATION VIA RE-HEATER. PRESSURE CONTROLLER (39-PIC-9503A) SHALL BE SELECTED DURING TANK WASHING OPERATION USING WASH WATER.
16. SLOPE FOR PSV OUTLET LINE CANNOT BE PROVIDED DUE TO LAYOUT CONSTRAINTS. LINE PROVIDED WITH INSULATION AND HEAT TRACING TO PREVENT BLOCKAGE.
17. VALVE TO BE INSTALLED AT HIGH POINT.
18. CLOSED DRAINS HEADER TO HAVE NO POCKETS AFTER PSV OUTLET LINE JOINS THE DRAINS HEADER.
19. FLANGE IS 300# IN AC51 PIPE CLASS.
20. EQUIPMENT TRIM: 39-HA-901A: N9: 2"-OT-41-8524-BC50-01 / 39-HA-901B; N9: 2"-OT-41-8544-BC50-01.
21. HIGH ALARM ON 39-FIT-9503A CLOSES 39-HV-9511 / 39-HV-9531.
22. HIGH HIGH ALARM ON 39-FIT-9503B TRIP CARGO PUMPS.

GENERAL NOTES

1. THE HEATERS TYPE HAS BEEN CHANGED TO SHELL & TUBE.
2. ALL INSTRUMENTATION HAS BEEN UPDATED.
3. TAG NUMBERS UPDATED.
4. REVISION P04 REFERS TO DCN-HJ-0862-1.
5. REVISION P05 REFERS TO DCN-HJ-1026-1. P05 DCN HJ-1026-1

HOLDS

1. DELETED.
2. DELETED.
3. DELETED.
4. DELETED.
5. DELETED.
6. DELETED.

Rev.	Date	Reason for Issue	Eng. by	Chk.	Appr.
P05	20.12.24	RE-ISSUED FOR CONSTRUCTION	AR	MSW	BØK
P04	23.10.24	RE-ISSUED FOR CONSTRUCTION	AR	MSW	BØK
P03	28.05.24	RE-ISSUED FOR CONSTRUCTION	AR	MSW	BØK
P02	17.11.23	ISSUED FOR CONSTRUCTION	AR	MSW	BØK
P01	24.05.23	ISSUED FOR HAZOP	MSW	MR	BØK

Vessel: ROSEBANK DEVELOPMENT - PJR FPSO

Document Title: PIPING AND INSTRUMENTATION DIAGRAM CRUDE OIL DECK RE-HEATERS (39-HA-901 A/B)

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